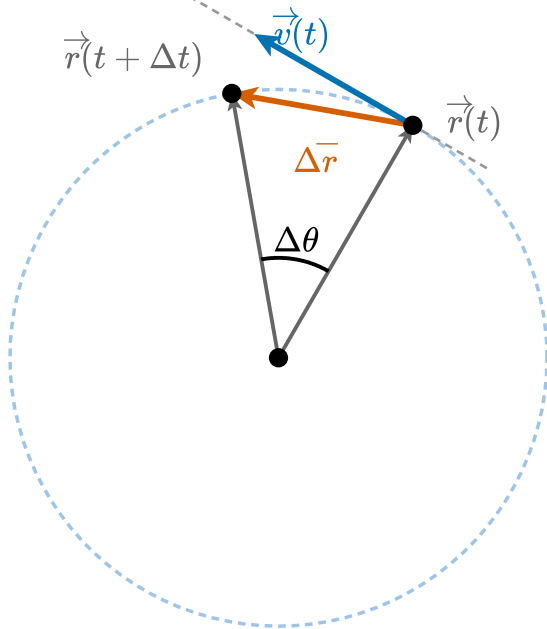


tangent line



$$\|\Delta \vec{r}\| = 2r \sin \frac{\Delta \theta}{2} \approx r \Delta \theta$$